

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549
FORM 10-K

(Mark One)

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the fiscal year ended December 31, 2024
OR

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the transition period from _____ to _____
Commission File Number: 001-34756

Tesla, Inc.

(Exact name of registrant as specified in its charter)

Texas
(State or other jurisdiction of
incorporation or organization)

91-2197729
(I.R.S. Employer
Identification No.)

1 Tesla Road
Austin, Texas
(Address of principal executive offices)

78725
(Zip Code)

(512) 516-8177
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common stock	TSLA	The Nasdaq Global Select Market

Securities registered pursuant to Section 12(g) of the Act:

None

Indicate by check mark whether the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☐ No ☒

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 ("Exchange Act") during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act:

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
Emerging growth company	☐		

TESLA, INC.

ANNUAL REPORT ON FORM 10-K FOR THE YEAR ENDED DECEMBER 31, 2024

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Management Opportunities, Challenges and Uncertainties and 2025 Outlook

Automotive—Production

The following is a summary of the status of production of each of our announced vehicle models in production and under development, as of the date of this Annual Report on Form 10-K:

Region	Vehicle Model(s)	Production Status
California	Model S / Model X	Active
	Model 3 / Model Y	Active
Shanghai	Model 3 / Model Y	Active
Berlin	Model Y	Active
Texas	Model Y	Active
	Cybertruck	Active
	Cybercab	In development
Nevada	Tesla Semi	Pilot production
TBD	Roadster	In development

We are focused on growing our manufacturing capacity, which includes capacity for manufacturing newer vehicle models such as our Cybertruck, Tesla Semi and future vehicles utilizing aspects of our next generation platform, and ramping the production at our Gigafactories to their installed production capacities as well as increasing production rate and efficiency at our current factories. The next phase of production growth will depend on the continued ramp at our factories and be initiated by advances in autonomy and the introduction of new products, including those built on our next generation vehicle platform, as well as our ability to add to our available sources of battery cell supply by manufacturing our own cells that we are developing to have high-volume output, lower capital and production costs and longer range. Our goals are to improve vehicle performance, decrease production costs and increase affordability and customer awareness.

These plans are subject to uncertainties inherent in establishing and ramping manufacturing operations, which may be exacerbated by new product and manufacturing technologies we introduce, the number of concurrent international projects, any industry-wide component constraints, labor shortages and any future impact from events outside of our control. For example, during the first quarter of 2024, we experienced a sequential decline in production volumes partially caused by the early phase of the production ramp of the updated Model 3 at our Fremont factory, and factory shutdowns at Gigafactory Berlin-Brandenburg resulting from shipping diversions caused by the Red Sea conflict and an arson attack. In the first quarter of 2025, as we launch our New Model Y worldwide, we may similarly experience delays or declines in production volumes due to simultaneous manufacturing ramps in facilities on three continents. Moreover, we have set ambitious technological targets with our plans for battery cells as well as for iterative manufacturing and design improvements for our vehicles with each new factory.

Automotive—Demand, Sales, Deliveries and Infrastructure

Our cost reduction efforts, cost innovation strategies, and additional localized procurement and manufacturing are key to our vehicles' affordability and have allowed us to competitively price our vehicles. We will also continue to generate demand by improving our vehicles' performance and functionality, including through product offerings and features based on artificial intelligence such as Autopilot, FSD (Supervised), and other software, and delivering new vehicles and vehicle options, such as our launch of the updated Model 3 in 2024, and the New Model Y in the first quarter of 2025. In addition, we have been increasing awareness, and expanding our vehicle financing programs, including attractive leasing terms for our customers.

Furthermore, significant judgment is required in evaluating our tax positions. In the ordinary course of business, there are many transactions and calculations for which the ultimate tax settlement is uncertain. As a result, we recognize the effect of this uncertainty on our tax attributes or taxes payable based on our estimates of the eventual outcome. These effects are recognized when, despite our belief that our tax return positions are supportable, we believe that it is more likely than not that some of those positions may not be fully sustained upon review by tax authorities. We are required to file income tax returns in the U.S. and various foreign jurisdictions, which requires us to interpret the applicable tax laws and regulations in effect in such jurisdictions. Such returns are subject to audit by the various federal, state and foreign taxing authorities, who may disagree with respect to our tax positions. We believe that our consideration is adequate for all open audit years based on our assessment of many factors, including past experience and interpretations of tax law. We review and update our estimates in light of changing facts and circumstances, such as the closing of a tax audit, the lapse of a statute of limitations or a change in estimate. To the extent that the final tax outcome of these matters differs from our expectations, such differences may impact income tax expense in the period in which such determination is made.

Results of Operations

Revenues

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Automotive sales	\$ 72,480	\$ 78,509	\$ 67,210	\$ (6,029)	(8)%	\$ 11,299	17 %
Automotive regulatory credits	2,763	1,790	1,776	973	54 %	14	1 %
Automotive leasing	1,827	2,120	2,476	(293)	(14)%	(356)	(14)%
Total automotive revenues	77,070	82,419	71,462	(5,349)	(6)%	10,957	15 %
Services and other	10,534	8,319	6,091	2,215	27 %	2,228	37 %
Total automotive & services and other segment revenue	87,604	90,738	77,553	(3,134)	(3)%	13,185	17 %
Energy generation and storage segment revenue	10,086	6,035	3,909	4,051	67 %	2,126	54 %
Total revenues	\$ 97,690	\$ 96,773	\$ 81,462	\$ 917	1 %	\$ 15,311	19 %

Automotive & Services and Other Segment

Automotive sales revenue includes revenues related to cash and financing deliveries of new vehicles, including access to our FSD (Supervised) features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. These deliveries are vehicles that are not subject to lease accounting.

Automotive regulatory credits includes sales of regulatory credits to other automotive manufacturers. Our revenue from automotive regulatory credits is directly related to our new vehicle production, sales and pricing negotiated with our customers. We monetize them proactively as new vehicles are sold based on standing arrangements with buyers of such credits, typically as close as possible to the production and delivery of the vehicle or changes in regulation impacting the credits.

Automotive leasing revenue includes the amortization of revenue for vehicles under direct operating lease agreements. Additionally, automotive leasing revenue includes direct sales-type leasing programs where we recognize all revenue associated with the sales-type lease upon delivery to the customer.

Services and other revenue consists of sales of used vehicles, non-warranty maintenance services and collision, part sales, paid Supercharging, insurance services revenue and retail merchandise sales.

2024 compared to 2023

Automotive sales revenue decreased \$6.03 billion, or 8%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to lower average selling price on our vehicles driven by overall price reductions and attractive financing options provided in 2024 as well as mix. Additionally, there was a decrease of approximately 22,000 combined Model 3 and Model Y cash deliveries. The decreases were partially offset by an increase of approximately 19,000 deliveries of other models primarily due to our production ramp of Cybertruck. Additionally, we recognized \$596 million of FSD (Supervised) revenue due to release of certain features in 2024.

Automotive regulatory credits revenue increased \$973 million, or 54%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, driven by demand for credits in North America as other automobile manufacturers scale back on their battery electric vehicle plans.

Automotive leasing revenue decreased \$293 million, or 14%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to lower direct sales-type leasing deliveries as we have shifted towards providing leasing options through commercial banking partner programs that allow for us to recognize upfront revenue in automotive sales and a decrease in lease buyouts.

Services and other revenue increased \$2.22 billion, or 27%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to increases in sales of used vehicles, non-warranty maintenance services and collision revenue, paid Supercharging revenue, insurance services revenue and part sales revenue.

Energy Generation and Storage Segment

Energy generation and storage revenue includes sales and leasing of solar energy generation and energy storage products, financing of solar energy generation products, services related to such products and sales of solar energy systems incentives.

2024 compared to 2023

Energy generation and storage revenue increased \$4.05 billion, or 67%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to a 16.7 GWh increase in Megapack and Powerwall deployments compared to the prior year.

Cost of Revenues and Gross Margin

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Cost of revenues							
Automotive sales	\$ 61,870	\$ 65,121	\$ 49,599	\$ (3,251)	(5)%	\$ 15,522	31 %
Automotive leasing	1,003	1,268	1,509	(265)	(21)%	(241)	(16)%
Total automotive cost of revenues	62,873	66,389	51,108	(3,516)	(5)%	15,281	30 %
Services and other	9,921	7,830	5,880	2,091	27 %	1,950	33 %
Total automotive & services and other segment cost of revenues	72,794	74,219	56,988	(1,425)	(2)%	17,231	30 %
Energy generation and storage segment	7,446	4,894	3,621	2,552	52 %	1,273	35 %
Total cost of revenues	\$ 80,240	\$ 79,113	\$ 60,609	\$ 1,127	1 %	\$ 18,504	31 %
Gross profit total automotive							
	\$ 14,197	\$ 16,030	\$ 20,354				
Gross margin total automotive							
	18.4 %	19.4 %	28.5 %				
Gross profit total automotive & services and other segment							
	\$ 14,810	\$ 16,519	\$ 20,565				
Gross margin total automotive & services and other segment							
	16.9 %	18.2 %	26.5 %				
Gross profit energy generation and storage segment							
	\$ 2,640	\$ 1,141	\$ 288				
Gross margin energy generation and storage segment							
	26.2 %	18.9 %	7.4 %				
Total gross profit							
	\$ 17,450	\$ 17,660	\$ 20,853				
Total gross margin							
	17.9 %	18.2 %	25.6 %				

Gross margin for energy generation and storage increased from 18.9% to 26.2% in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to margin improvements for our energy storage products driven by cost reductions, including benefits from IRA manufacturing credits, and a higher proportion of our storage business, which operated at a higher gross margin, within the segment as compared to the prior periods.

Research and Development Expense

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Research and development	\$ 4,540	\$ 3,969	\$ 3,075	\$ 571	14 %	\$ 894	29 %
As a percentage of revenues	5 %	4 %	4 %				

Research and development (“R&D”) expenses consist primarily of personnel costs for our teams in engineering and research, manufacturing engineering and manufacturing test organizations, prototyping expense, contract and professional services and amortized equipment expense.

R&D expenses increased \$571 million, or 14%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023. The overall increases were primarily driven by additional costs year over year related to AI programs. R&D expenses as a percentage of revenue increased from 4% to 5% in the year ended December 31, 2024 as compared to the year ended December 31, 2023 as we continue to expand our product roadmap and technologies.

Selling, General and Administrative Expense

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Selling, general and administrative	\$ 5,150	\$ 4,800	\$ 3,946	\$ 350	7 %	\$ 854	22 %
As a percentage of revenues	5 %	5 %	5 %				

Selling, general and administrative (“SG&A”) expenses generally consist of personnel and facilities costs related to our stores, marketing, sales, executive, finance, human resources, information technology and legal organizations, as well as fees for professional and contract services and litigation settlements.

SG&A expenses increased \$350 million, or 7%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023 driven by a \$171 million increase in facilities related expenses, a \$115 million increase in employee and labor costs, including professional services and a \$57 million increase in marketing expenses.

Restructuring and Other

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Restructuring and other	\$ 684	\$ —	\$ 176	\$ 684	Not meaningful	\$ (176)	(100)%

In the second quarter of 2024, we initiated and substantially completed certain restructuring actions to reduce costs and improve efficiency. As a result, we recognized \$583 million of employee termination expenses in Restructuring and other in our consolidated income statement. These expenses were substantially paid with an immaterial accrual remaining in Accrued liabilities and other in our consolidated balance sheet as of December 31, 2024.

Interest Income

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Interest income	\$ 1,569	\$ 1,066	\$ 297	\$ 503	47 %	\$ 769	259 %

Interest income increased \$503 million, or 47%, in the year ended December 31, 2024 as compared to the year ended December 31, 2023, primarily due to higher interest earned on our cash and cash equivalents and short-term investments compared to the prior year due to an increase in our portfolio balance and a higher weighted average interest rate.

Other Income (Expense), Net

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Other income (expense), net	\$ 695	\$ 172	\$ (43)	\$ 523	304%	\$ 215	Not meaningful

Other income (expense), net, consists of foreign exchange gains and losses related to our foreign currency-denominated monetary assets and liabilities and fair value remeasurements of our digital assets following the adoption of ASU 2023-08 effective January 1, 2024. See Note 2, *Summary of Significant Accounting Policies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Other income (expense), net, changed favorably by \$523 million in the year ended December 31, 2024 as compared to the year ended December 31, 2023 primarily due to remeasurement of our bitcoin digital assets to fair value in 2024 (see above), partially offset by unfavorable fluctuations in foreign currency exchange rates on our intercompany balances. As our intercompany balances are significant in nature and we do not typically hedge foreign currency risk, we can experience significant fluctuations in foreign currency exchange rate gains and losses from period to period.

Provision for (Benefit from) Income Taxes

(Dollars in millions)	Year Ended December 31,			2024 vs. 2023 Change		2023 vs. 2022 Change	
	2024	2023	2022	\$	%	\$	%
Provision for (benefit from) income taxes	\$ 1,837	\$ (5,001)	\$ 1,132	\$ 6,838	Not meaningful	\$ (6,133)	Not meaningful
Effective tax rate	20 %	(50)%	8 %				

Our provision for (benefit from) income taxes changed by \$6.84 billion in the year ended December 31, 2024 as compared to the year ended December 31, 2023. Our effective tax rate changed to an expense of 20% in the year ended December 31, 2024 from a benefit of 50% in the year ended December 31, 2023. These changes are primarily due to the impact of releasing the valuation allowance on our U.S. deferred tax assets in the fourth quarter of 2023.

We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. In the fourth quarter of 2023, based on the relevant weight of positive and negative evidence, including the amount of our taxable income in recent years which was objective and verifiable, and consideration of our expected future taxable earnings, we concluded that it is more likely than not that most of our U.S. federal and certain state deferred tax assets are realizable and released \$6.54 billion of our valuation allowance. As of December 31, 2024, we continue to conclude that it is more likely than not that our deferred tax assets are realizable, except for our California deferred tax assets, U.S. foreign tax credits and certain foreign operating losses.

See Note 13, *Income Taxes*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Liquidity and Capital Resources

We expect to continue to generate net positive operating cash flow. The cash we generate from our core operations enables us to fund ongoing operations and production, our research and development projects for new products and technologies including our proprietary battery cells, additional manufacturing ramps at existing manufacturing facilities, the construction of future factories, and the continued expansion of our retail and service locations, body shops, Mobile Service fleet, Supercharger, including to support NACS, energy product installation capabilities and autonomy and other artificial intelligence enabled products.

In addition, because a large portion of our future expenditures will be to fund our growth, we expect that if needed we will be able to adjust our capital and operating expenditures by operating segment. For example, if our near-term manufacturing operations decrease in scale or ramp more slowly than expected, including due to global economic or business conditions, we may choose to correspondingly slow the pace of our capital expenditures. Finally, we continually evaluate our cash needs and may decide it is best to raise additional capital or seek alternative financing sources to fund the rapid growth of our business, including through drawdowns on existing or new debt facilities or financing funds. Conversely, we may also from time to time determine that it is in our best interests to voluntarily repay certain indebtedness early.

Accordingly, we believe that our current sources of funds will provide us with adequate liquidity during the 12-month period following December 31, 2024, as well as in the long-term.

See the sections below for more details regarding the material requirements for cash in our business and our sources of liquidity to meet such needs.

Material Cash Requirements

From time to time in the ordinary course of business, we enter into agreements with vendors for the purchase of components and raw materials to be used in the manufacture of our products. However, due to contractual terms, variability in the precise growth curves of our development and production ramps, and opportunities to renegotiate pricing, we generally do not have binding and enforceable purchase orders under such contracts beyond the short-term, and the timing and magnitude of purchase orders beyond such period is difficult to accurately project.

As discussed in and subject to the considerations referenced in Part II, Item 7, Management's Discussion and Analysis of Financial Condition and Results of Operations—Management Opportunities, Challenges and Uncertainties and 2025 Outlook—Cash Flow and Capital Expenditure Trends in this Annual Report on Form 10-K, we currently expect our capital expenditures to support our projects globally to exceed \$11.00 billion in 2025 and in each of the following two fiscal years.

As of December 31, 2024, we and our subsidiaries had outstanding \$7.91 billion in aggregate principal amount of indebtedness, of which \$2.35 billion is current. As of December 31, 2024, our total minimum lease payments was \$7.05 billion, of which \$1.19 billion is due in the succeeding 12 months. For details regarding our indebtedness and lease obligations, refer to Note 10, *Debt*, and Note 11, *Leases*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

Sources and Conditions of Liquidity

Our sources to fund our material cash requirements are predominantly from our deliveries and servicing of new and used vehicles, sales and installations of our energy storage products, interest income, and proceeds from debt facilities and equity offerings, when applicable.

As of December 31, 2024, we had \$16.14 billion and \$20.42 billion of cash and cash equivalents and short-term investments, respectively. Balances held in foreign currencies had a U.S. dollar equivalent of \$2.90 billion and consisted primarily of euros and Chinese yuan. We had \$5.00 billion of unused committed credit amounts as of December 31, 2024. For details regarding our indebtedness, refer to Note 10, *Debt*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

We continue adapting our strategy to meet our liquidity and risk objectives, such as investing in U.S. government securities and other investments, invest in autonomy, do more vertical integration, expand our product roadmap and provide financing options to our customers.

Summary of Cash Flows

(Dollars in millions)	Year Ended December 31,		
	2024	2023	2022
Net cash provided by operating activities	\$ 14,923	\$ 13,256	\$ 14,724
Net cash used in investing activities	\$ (18,787)	\$ (15,584)	\$ (11,973)
Net cash provided by (used in) financing activities	\$ 3,853	\$ 2,589	\$ (3,527)

Cash Flows from Operating Activities

Our cash flows from operating activities are significantly affected by our cash investments to support the growth of our business in areas such as research and development and selling, general and administrative and working capital. Our operating cash inflows include cash from vehicle sales and related servicing, customer lease and financing payments, customer deposits, cash from sales of regulatory credits and energy generation and storage products, and interest income on our cash and investments portfolio. These cash inflows are offset by our payments to suppliers for production materials and parts used in our manufacturing process, operating expenses, operating lease payments and interest payments on our financings.

Tesla, Inc.
Consolidated Balance Sheets
(in millions, except per share data)

	December 31, 2024	December 31, 2023
Assets		
Current assets		
Cash and cash equivalents	\$ 16,139	\$ 16,398
Short-term investments	20,424	12,696
Accounts receivable, net	4,418	3,508
Inventory	12,017	13,626
Prepaid expenses and other current assets	5,362	3,388
Total current assets	58,360	49,616
Operating lease vehicles, net	5,581	5,989
Solar energy systems, net	4,924	5,229
Property, plant and equipment, net	35,836	29,725
Operating lease right-of-use assets	5,160	4,180
Digital assets, net	1,076	184
Intangible assets, net	150	178
Goodwill	244	253
Deferred tax assets	6,524	6,733
Other non-current assets	4,215	4,531
Total assets	\$ 122,070	\$ 106,618
Liabilities		
Current liabilities		
Accounts payable	\$ 12,474	\$ 14,431
Accrued liabilities and other	10,723	9,080
Deferred revenue	3,168	2,864
Current portion of debt and finance leases	2,456	2,373
Total current liabilities	28,821	28,748
Debt and finance leases, net of current portion	5,757	2,857
Deferred revenue, net of current portion	3,317	3,251
Other long-term liabilities	10,495	8,153
Total liabilities	48,390	43,009
Commitments and contingencies (Note 14)		
Redeemable noncontrolling interests in subsidiaries	63	242
Equity		
Stockholders' equity		
Preferred stock; \$0.001 par value; 100 shares authorized; no shares issued and outstanding	—	—
Common stock; \$0.001 par value; 6,000 shares authorized; 3,216 and 3,185 shares issued and outstanding as of December 31, 2024 and December 31, 2023, respectively	3	3
Additional paid-in capital	38,371	34,892
Accumulated other comprehensive loss	(670)	(143)
Retained earnings	35,209	27,882
Total stockholders' equity	72,913	62,634
Noncontrolling interests in subsidiaries	704	733
Total liabilities and equity	\$ 122,070	\$ 106,618

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.
Consolidated Statements of Operations
(in millions, except per share data)

	Year Ended December 31,		
	2024	2023	2022
Revenues			
Automotive sales	\$ 72,480	\$ 78,509	\$ 67,210
Automotive regulatory credits	2,763	1,790	1,776
Automotive leasing	1,827	2,120	2,476
Total automotive revenues	77,070	82,419	71,462
Energy generation and storage	10,086	6,035	3,909
Services and other	10,534	8,319	6,091
Total revenues	97,690	96,773	81,462
Cost of revenues			
Automotive sales	61,870	65,121	49,599
Automotive leasing	1,003	1,268	1,509
Total automotive cost of revenues	62,873	66,389	51,108
Energy generation and storage	7,446	4,894	3,621
Services and other	9,921	7,830	5,880
Total cost of revenues	80,240	79,113	60,609
Gross profit	17,450	17,660	20,853
Operating expenses			
Research and development	4,540	3,969	3,075
Selling, general and administrative	5,150	4,800	3,946
Restructuring and other	684	—	176
Total operating expenses	10,374	8,769	7,197
Income from operations	7,076	8,891	13,656
Interest income	1,569	1,066	297
Interest expense	(350)	(156)	(191)
Other income (expense), net	695	172	(43)
Income before income taxes	8,990	9,973	13,719
Provision for (benefit from) income taxes	1,837	(5,001)	1,132
Net income	7,153	14,974	12,587
Net income (loss) attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries	62	(23)	31
Net income attributable to common stockholders	\$ 7,091	\$ 14,997	\$ 12,556
Net income per share of common stock attributable to common stockholders			
Basic	\$ 2.23	\$ 4.73	\$ 4.02
Diluted	\$ 2.04	\$ 4.30	\$ 3.62
Weighted average shares used in computing net income per share of common stock			
Basic	3,197	3,174	3,130
Diluted	3,498	3,485	3,475

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.
Consolidated Statements of Comprehensive Income
(in millions)

	Year Ended December 31,		
	2024	2023	2022
Net income	\$ 7,153	\$ 14,974	\$ 12,587
Other comprehensive income (loss):			
Foreign currency translation adjustment	(539)	198	(392)
Unrealized net gain (loss) on investments, net of tax	12	16	(23)
Net loss realized and included in net income	—	4	—
Comprehensive income	6,626	15,192	12,172
Less: Comprehensive income (loss) attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries	62	(23)	31
Comprehensive income attributable to common stockholders	\$ 6,564	\$ 15,215	\$ 12,141

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Redeemable Noncontrolling Interests and Equity
(in millions)

	Redeemable Noncontrolling Interests	Common Stock		Additional Paid-In Capital	Accumulated Other Comprehensive Income (Loss)	Retained Earnings	Total Stockholders' Equity	Noncontrolling Interests in Subsidiaries	Total Equity
		Shares	Amount						
Balance as of December 31, 2021	\$ 568	3,100	\$ 3	\$ 29,803	\$ 54	\$ 329	\$ 30,189	\$ 826	\$ 31,015
Settlements of warrants	—	37	—	—	—	—	—	—	—
Issuance of common stock for equity incentive awards	—	27	—	541	—	—	541	—	541
Stock-based compensation	—	—	—	1,806	—	—	1,806	—	1,806
Distributions to noncontrolling interests	(46)	—	—	—	—	—	—	(113)	(113)
Buy-outs of noncontrolling interests	(11)	—	—	27	—	—	27	(61)	(34)
Net (loss) income	(102)	—	—	—	—	12,556	12,556	133	12,689
Other comprehensive loss	—	—	—	—	(415)	—	(415)	—	(415)
Balance as of December 31, 2022	\$ 409	3,164	\$ 3	\$ 32,177	\$ (361)	\$ 12,885	\$ 44,704	\$ 785	\$ 45,489
Issuance of common stock for equity incentive awards	—	21	—	700	—	—	700	—	700
Stock-based compensation	—	—	—	2,013	—	—	2,013	—	2,013
Distributions to noncontrolling interests	(32)	—	—	—	—	—	—	(108)	(108)
Buy-outs of noncontrolling interests	(39)	—	—	2	—	—	2	(17)	(15)
Net (loss) income	(96)	—	—	—	—	14,997	14,997	73	15,070
Other comprehensive income	—	—	—	—	218	—	218	—	218
Balance as of December 31, 2023	\$ 242	3,185	\$ 3	\$ 34,892	\$ (143)	\$ 27,882	\$ 62,634	\$ 733	\$ 63,367
Adjustments for prior periods from adopting ASU 2023-08, net of tax	—	—	—	—	—	236	236	—	236
Settlements of warrants	—	11	—	—	—	—	—	—	—
Issuance of common stock for equity incentive awards	—	20	—	1,241	—	—	1,241	—	1,241
Stock-based compensation	—	—	—	2,199	—	—	2,199	—	2,199
Distributions to noncontrolling interests	(13)	—	—	—	—	—	—	(85)	(85)
Buy-outs of noncontrolling interests	(172)	—	—	39	—	—	39	—	39
Net income	6	—	—	—	—	7,091	7,091	56	7,147
Other comprehensive loss	—	—	—	—	(527)	—	(527)	—	(527)
Balance as of December 31, 2024	\$ 63	3,216	\$ 3	\$ 38,371	\$ (670)	\$ 35,209	\$ 72,913	\$ 704	\$ 73,617

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.
Consolidated Statements of Cash Flows
(in millions)

	Year Ended December 31,		
	2024	2023	2022
Cash Flows from Operating Activities			
Net income	\$ 7,153	\$ 14,974	\$ 12,587
Adjustments to reconcile net income to net cash provided by operating activities:			
Depreciation, amortization and impairment	5,368	4,667	3,747
Stock-based compensation	1,999	1,812	1,560
Inventory and purchase commitments write-downs	335	463	177
Foreign currency transaction net unrealized (gain) loss	(73)	(144)	81
Deferred income taxes	477	(6,349)	(196)
Non-cash interest and other operating activities	172	81	340
Digital assets (gain) loss, net	(589)	—	140
Changes in operating assets and liabilities:			
Accounts receivable	(1,083)	(586)	(1,124)
Inventory	937	(1,195)	(6,465)
Operating lease vehicles	(590)	(1,952)	(1,570)
Prepaid expenses and other assets	(3,273)	(2,652)	(3,713)
Accounts payable, accrued and other liabilities	3,588	2,605	8,029
Deferred revenue	502	1,532	1,131
Net cash provided by operating activities	14,923	13,256	14,724
Cash Flows from Investing Activities			
Purchases of property and equipment excluding finance leases, net of sales	(11,339)	(8,898)	(7,158)
Purchases of solar energy systems, net of sales	(3)	(1)	(5)
Proceeds from sales of digital assets	—	—	936
Purchase of intangible assets	—	—	(9)
Purchases of investments	(35,955)	(19,112)	(5,835)
Proceeds from maturities of investments	28,310	12,353	22
Proceeds from sales of investments	200	138	—
Receipt of government grants	—	—	76
Business combinations, net of cash acquired	—	(64)	—
Net cash used in investing activities	(18,787)	(15,584)	(11,973)
Cash Flows from Financing Activities			
Proceeds from issuances of debt	5,744	3,931	—
Repayments of debt	(2,500)	(1,351)	(3,364)
Proceeds from exercises of stock options and other stock issuances	1,241	700	541
Principal payments on finance leases	(381)	(464)	(502)
Debt issuance costs	(14)	(29)	—
Distributions paid to noncontrolling interests in subsidiaries	(104)	(144)	(157)
Payments for buy-outs of noncontrolling interests in subsidiaries	(133)	(54)	(45)
Net cash provided by (used in) financing activities	3,853	2,589	(3,527)
Effect of exchange rate changes on cash and cash equivalents and restricted cash	(141)	4	(444)
Net (decrease) increase in cash and cash equivalents and restricted cash	(152)	265	(1,220)
Cash and cash equivalents and restricted cash, beginning of period	17,189	16,924	18,144
Cash and cash equivalents and restricted cash, end of period	\$ 17,037	\$ 17,189	\$ 16,924
Supplemental Non-Cash Investing and Financing Activities			
Acquisitions of property and equipment included in liabilities	\$ 1,410	\$ 2,272	\$ 2,148
Supplemental Disclosures			
Cash paid during the period for interest	\$ 277	\$ 126	\$ 152
Cash paid during the period for income taxes, net of refunds	\$ 1,331	\$ 1,119	\$ 1,203

The accompanying notes are an integral part of these consolidated financial statements.

Revenue Recognition

Revenue by source

The following table disaggregates our revenue by major source (in millions):

	Year Ended December 31,		
	2024	2023	2022
Automotive sales	\$ 72,480	\$ 78,509	\$ 67,210
Automotive regulatory credits	2,763	1,790	1,776
Energy generation and storage sales	9,564	5,515	3,376
Services and other	10,534	8,319	6,091
Total revenues from sales and services	95,341	94,133	78,453
Automotive leasing	1,827	2,120	2,476
Energy generation and storage leasing	522	520	533
Total revenues	\$ 97,690	\$ 96,773	\$ 81,462

Automotive Segment

Automotive Sales

Automotive sales revenue includes revenues related to cash and financing deliveries of new vehicles, and specific other features and services that meet the definition of a performance obligation under ASC 606, including access to our FSD(Supervised) features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. We recognize revenue on automotive sales upon delivery to the customer, which is when the control of a vehicle transfers. Payments are typically received at the point control transfers or in accordance with payment terms customary to the business, except sales we finance for which payments are collected over the contractual loan term. We also recognize a sales return reserve based on historical experience plus consideration for expected future market values, when we offer resale value guarantees or similar buyback terms. Other features and services such as access to our internet connectivity, unlimited free Supercharging and over-the-air software updates are provisioned upon control transfer of a vehicle and recognized over time on a straight-line basis as we have a stand-ready obligation to deliver such services to the customer. Other limited free Supercharging incentives are recognized based on actual usage or expiration, whichever is earlier. We recognize revenue related to these other features and services over the performance period, which is generally the expected ownership life of the vehicle. Revenue related to FSD (Supervised) features is recognized when functionality is delivered to the customer and their ongoing maintenance is recognized over time. For our obligations related to automotive sales, we estimate standalone selling price by considering costs used to develop and deliver the service, third-party pricing of similar options and other information that may be available.

Any fees that are paid or payable by us to a customer's lender when we arrange the financing are recognized upfront as an offset against automotive sales revenue. Costs to obtain a contract mainly relate to commissions for the sale of vehicles. As our contract costs related to automotive sales are typically fulfilled within one year, the costs to obtain a contract are expensed as incurred. Amounts billed to customers related to shipping and handling are classified as automotive sales revenue, and we have elected to recognize the cost for freight and shipping when control over vehicles, parts or accessories have transferred to the customer as an expense in cost of automotive sales revenue. Our policy is to exclude taxes collected from a customer from the transaction price of automotive contracts.

We offer resale value guarantees to our commercial banking partners in connection with certain vehicle leasing programs. Under these programs, we originate the lease with our end customer and immediately transfer the lease and the underlying vehicle to our commercial banking partner, with the transaction being accounted for as a sale under ASC 606.

We receive upfront payment for the vehicle, do not bear casualty and credit risks during the lease term, and we provide a guarantee capped to a limit if they are unable to sell the vehicle at or above the vehicle's contractual or determined residual value at the end of the lease term. We estimate a guarantee liability in accordance with ASC 460, *Guarantees* and record it within other liabilities on our consolidated balance sheet. On a quarterly basis, we assess the estimated market value of vehicles sold under these programs to determine whether there have been changes to the amount of expected resale value guarantee liabilities. As we accumulate more data related to the resale values of our vehicles or as market conditions change, there may be material changes to their estimated values. The total recorded guarantee liabilities on vehicles sold under these programs were immaterial as of December 31, 2024 and 2023. Our maximum exposure on the guarantees we provide if they are unable to sell the vehicle at or above the vehicle's contractual residual value at the end of the lease term was \$1.45 billion and \$166 million as of December 31, 2024 and 2023, respectively.

Deferred revenue related to the access to our FSD (Supervised) features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates primarily on automotive sales consisted of the following (in millions):

	Year Ended December 31,	
	2024	2023
Deferred revenue — beginning of period	\$ 3,536	\$ 2,913
Additions	1,343	1,201
Net changes in liability for pre-existing contracts, including foreign exchange impact	(92)	17
Revenue recognized	(1,188)	(595)
Deferred revenue — end of period	\$ 3,599	\$ 3,536

Deferred revenue is equivalent to the total transaction price allocated to the performance obligations that are unsatisfied, or partially unsatisfied, as of the balance sheet date. Revenue recognized from the deferred revenue balance as of December 31, 2023 and 2022 was \$872 million and \$469 million for the years ended December 31, 2024 and 2023, respectively. Of the total deferred revenue balance as of December 31, 2024, we expect to recognize \$840 million of revenue in the next 12 months. The remaining balance will be recognized at the time of transfer of control of the product or over the performance period as discussed above in Automotive Sales.

We have financing receivables on our consolidated balance sheets related to loans we provide for financing our automotive deliveries. As of December 31, 2024 and 2023, we had current net financing receivables of \$247 million and \$242 million, respectively, in Accounts receivable, net, and \$821 million and \$1.04 billion, respectively, in Other non-current assets for the long-term portion.

Automotive Regulatory Credits

We earn tradable credits in the operation of our automotive business under various regulations related to ZEVs, greenhouse gas, fuel economy and clean fuel. We sell these credits to other regulated entities who can use the credits to comply with emission standards and other regulatory requirements.

Payments for automotive regulatory credits are typically received at the point control transfers to the customer, or in accordance with payment terms customary to the business. We recognize revenue on the sale of automotive regulatory credits, which have negligible incremental costs associated with them, at the time control of the regulatory credits is transferred to the purchasing party. Deferred revenue related to sales of automotive regulatory credits was immaterial as of December 31, 2024 and 2023. Revenue recognized from the deferred revenue balance as of December 31, 2023 and 2022 was immaterial for the years ended December 31, 2024 and 2023. During the year ended December 31, 2022, we had also recognized \$288 million in revenue due to changes in regulation which entitled us to additional consideration for credits sold previously. We have elected the practical expedient to omit disclosure of the amount of the transaction price allocated to remaining performance obligations for contracts with an original expected contract length of one year or less. As of December 31, 2024, total transaction price allocated to performance obligations that were unsatisfied or partially unsatisfied for contracts with an original expected length of more than one year was \$4.68 billion. Of this amount, we expect to recognize \$863 million in the next 12 months and the rest over the remaining performance obligation period. Additionally, changes in regulations on automotive regulatory credits may significantly impact our remaining performance obligations and revenue to be recognized under these contracts.

We capitalize initial direct costs from the execution of agreements for solar energy systems and PPAs, which include the referral fees and sales commissions, as an element of solar energy systems, net, and subsequently amortize these costs over the term of the related agreements.

Cost of Revenues

Automotive Segment

Automotive Sales

Cost of automotive sales revenue includes direct and indirect materials, labor costs, manufacturing overhead, including depreciation costs of tooling and machinery, shipping and logistic costs, vehicle connectivity costs, FSD (Supervised) ongoing maintenance costs, electricity costs for our free Supercharging programs and reserves for estimated warranty expenses. Cost of automotive sales revenues also includes adjustments to warranty expense and charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of automotive sales revenue benefits from manufacturing credits earned, amounting to \$625 million and \$359 million for the years ended December 31, 2024 and 2023, respectively.

Automotive Leasing

Cost of automotive leasing revenue includes the depreciation of operating lease vehicles, cost of goods sold associated with direct sales-type leases and warranty expense related to leased vehicles.

Services and Other

Costs of services and other revenue includes cost of used vehicles including refurbishment costs, costs associated with providing non-warranty after-sales services, costs associated with our body shops and part sales, costs of paid Supercharging, costs to provide vehicle insurance and costs of retail merchandise sales.

Energy Generation and Storage Segment

Energy Generation and Storage

Cost of energy generation and storage revenue includes direct and indirect material and labor costs, manufacturing overhead, including depreciation costs of tooling and machinery, freight, warranty expense, and cost of servicing. Cost of energy generation and storage revenue also includes charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of energy generation and storage revenue benefits from manufacturing credits earned, amounting to \$756 million and \$115 million for the years ended December 31, 2024 and 2023, respectively. In agreements for solar energy systems and PPAs where we are the lessor, the cost of revenue is primarily comprised of depreciation of the cost of leased solar energy systems, maintenance costs associated with those systems and amortization of any initial direct costs.

Research and Development Costs

Research and development costs are expensed as incurred.

Income Taxes

We are subject to income taxes in the U.S. and in many foreign jurisdictions. Income taxes are computed using the asset and liability method, under which deferred tax assets and liabilities are determined based on the difference between the financial statement and tax bases of assets and liabilities using enacted tax rates in effect for the year in which the differences are expected to affect taxable income. Valuation allowances are established when necessary to reduce deferred tax assets to the amount expected to be realized.

Noncontrolling Interests and Redeemable Noncontrolling Interests

Noncontrolling interests and redeemable noncontrolling interests represent third-party interests in the net assets under certain funding arrangements, or funds, that we have entered into to finance the costs of solar energy systems and vehicles under operating leases. We have determined that the contractual provisions of the funds represent substantive profit-sharing arrangements. We have further determined that the methodology for calculating the noncontrolling interest and redeemable noncontrolling interest balances that reflects the substantive profit-sharing arrangements is a balance sheet approach using the hypothetical liquidation at book value ("HLBV") method. We, therefore, determine the amount of the noncontrolling interests and redeemable noncontrolling interests in the net assets of the funds at each balance sheet date using the HLBV method, which is presented on the consolidated balance sheet as noncontrolling interests in subsidiaries and redeemable noncontrolling interests in subsidiaries. Under the HLBV method, the amounts reported as noncontrolling interests and redeemable noncontrolling interests in the consolidated balance sheet represent the amounts the third parties would hypothetically receive at each balance sheet date under the liquidation provisions of the funds, assuming the net assets of the funds were liquidated at their recorded amounts determined in accordance with GAAP and with tax laws effective at the balance sheet date and distributed to the third parties. The third parties' interests in the results of operations of the funds are determined as the difference in the noncontrolling interest and redeemable noncontrolling interest balances in the consolidated balance sheets between the start and end of each reporting period, after taking into account any capital transactions between the funds and the third parties. However, the redeemable noncontrolling interest balance is at least equal to the redemption amount. The redeemable noncontrolling interest balance is presented as temporary equity in the mezzanine section of the consolidated balance sheet since these third parties have the right to redeem their interests in the funds for cash or other assets.

Net Income per Share of Common Stock Attributable to Common Stockholders

Basic net income per share of common stock attributable to common stockholders is calculated by dividing net income attributable to common stockholders by the weighted-average shares of common stock outstanding for the period. Potentially dilutive shares, which are based on the weighted-average shares of common stock underlying outstanding stock-based awards, warrants and convertible senior notes using the treasury stock method or the if-converted method, as applicable, are included when calculating diluted net income per share of common stock attributable to common stockholders when their effect is dilutive.

The following table presents the reconciliation of net income attributable to common stockholders to net income used in computing basic and diluted net income per share of common stock (in millions):

	Year Ended December 31,		
	2024	2023	2022
Net income attributable to common stockholders	\$ 7,091	\$ 14,997	\$ 12,556
Less: Buy-outs of noncontrolling interests	(39)	(2)	(27)
Net income used in computing basic net income per share of common stock	7,130	14,999	12,583
Less: Dilutive convertible debt	—	—	(1)
Net income used in computing diluted net income per share of common stock	\$ 7,130	\$ 14,999	\$ 12,584

The following table presents the reconciliation of basic to diluted weighted average shares used in computing net income per share of common stock attributable to common stockholders (in millions):

	Year Ended December 31,		
	2024	2023	2022
Weighted average shares used in computing net income per share of common stock, basic	3,197	3,174	3,130
Add:			
Stock-based awards	292	298	310
Convertible senior notes	1	2	3
Warrants	8	11	32
Weighted average shares used in computing net income per share of common stock, diluted	3,498	3,485	3,475

The following table presents the potentially dilutive shares that were excluded from the computation of diluted net income per share of common stock attributable to common stockholders, because their effect was anti-dilutive (in millions):

	Year Ended December 31,		
	2024	2023	2022
Stock-based awards	15	12	4

Cash and Cash Equivalents

All highly liquid investments with an original maturity of three months or less at the date of purchase are considered cash equivalents.

Restricted Cash

We maintain certain cash balances restricted as to withdrawal or use. Our restricted cash is comprised primarily of cash held to service certain payments under various secured debt facilities, cash held as collateral for sales to lease partners with a resale value guarantee and deposits held for our insurance services. We record restricted cash as other assets in the consolidated balance sheets and determine current or non-current classification based on the expected duration of the restriction.

Our total cash and cash equivalents and restricted cash, as presented in the consolidated statements of cash flows, was as follows (in millions):

	December 31, 2024	December 31, 2023	December 31, 2022
Cash and cash equivalents	\$ 16,139	\$ 16,398	\$ 16,253
Restricted cash included in prepaid expenses and other current assets	494	543	294
Restricted cash included in other non-current assets	404	248	377
Total as presented in the consolidated statements of cash flows	\$ 17,037	\$ 17,189	\$ 16,924

Investments

Our investments are all designated as available-for-sale and reported at estimated fair value, with unrealized gains and losses recorded in accumulated other comprehensive loss which is included within stockholders' equity. Available-for-sale marketable securities with maturities greater than three months at the date of purchase are included in short-term investments in our consolidated balance sheets. Interest, dividends, amortization and accretion of purchase premiums and discounts on these investments are included within Interest income in our consolidated statements of operations. Interest income on our short-term investments for the years ended December 31, 2024, 2023 and 2022 was \$763 million, \$388 million and an immaterial amount, respectively.

For periods prior to January 1, 2024, impairment losses were recognized within Restructuring and other in the consolidated statements of operations in the period in which the impairment was identified. Also for periods prior to January 1, 2024, gains were not recorded until realized upon sale(s), at which point they were presented net of any impairment losses for the same digital assets held within Restructuring and other. In determining the gain to be recognized upon sale, we calculate the difference between the sales price and carrying value of the digital assets sold immediately prior to sale.

See Note 3, *Digital Assets, Net*, for further information regarding digital assets.

Solar Energy Systems, Net

We are the lessor of solar energy systems, which are stated at cost less accumulated depreciation. Depreciation of solar energy systems is calculated using the straight-line method over the estimated useful lives of 30 to 35 years. The gross cost of solar energy systems as of December 31, 2024 and 2023 was \$6.79 billion and \$6.87 billion, respectively. Solar energy systems on the consolidated balance sheets are presented net of accumulated depreciation of \$1.86 billion and \$1.64 billion as of December 31, 2024 and 2023, respectively.

Property, Plant and Equipment, Net

Property, plant and equipment, net, including leasehold improvements, are recognized at cost less accumulated depreciation. Depreciation is generally computed using the straight-line method over the estimated useful lives of the respective assets, as follows:

Machinery, equipment, vehicles and office furniture	3 to 15 years
Tooling	4 to 7 years
Building and building improvements	15 to 30 years
Computer equipment and software	3 to 6 years
AI infrastructure	5 to 30 years

Leasehold improvements are depreciated on a straight-line basis over their estimated useful lives. AI infrastructure includes our owned data centers.

Upon the retirement or sale of our property, plant and equipment, the cost and associated accumulated depreciation are removed from the consolidated balance sheet, and the resulting gain or loss is reflected on the consolidated statement of operations. Maintenance and repair expenditures are expensed as incurred while major improvements that increase the functionality, output or expected life of an asset are capitalized and depreciated ratably over the identified useful life.

Long-Lived Assets Including Acquired Intangible Assets

We review our property, plant and equipment, solar energy systems, long-term prepayments and intangible assets for impairment whenever events or changes in circumstances indicate that the carrying amount of an asset (or asset group) may not be recoverable. We measure recoverability by comparing the carrying amount to the future undiscounted cash flows that the asset is expected to generate. If the asset is not recoverable, its carrying amount would be adjusted down to its fair value. For the years ended December 31, 2024, 2023 and 2022, we have recognized no material impairments of our long-lived assets.

Intangible assets with definite lives are amortized on a straight-line basis over their estimated useful lives, which range from seven to thirty years.

Goodwill

We assess goodwill for impairment annually in the fourth quarter, or more frequently if events or changes in circumstances indicate that it might be impaired, by comparing its carrying value to the reporting unit's fair value. For the years ended December 31, 2024, 2023, and 2022, we did not recognize any impairment of goodwill.

Capitalization of Software Costs

We capitalize costs incurred in the development of internal use software, during the application development stage to Property, plant and equipment, net on the consolidated balance sheets. Costs related to preliminary project activities and post-implementation activities are expensed as incurred. Such costs are amortized on a straight-line basis over their estimated useful life of three to five years.

Software development costs incurred in development of software to be sold, leased, or otherwise marketed, incurred subsequent to the establishment of technological feasibility and prior to the general availability of the software are capitalized when they are expected to become significant. Such costs are amortized over the estimated useful life of the applicable software once it is made generally available to our customers.

We evaluate the useful lives of these assets on an annual basis, and we test for impairment whenever events or changes in circumstances occur that could impact the recoverability of these assets. For the years ended December 31, 2024, 2023, and 2022, we have recognized no impairments of capitalized software costs.

Foreign Currency

We determine the functional and reporting currency of each of our international subsidiaries and their operating divisions based on the primary currency in which they operate. In cases where the functional currency is not the U.S. dollar, we recognize a cumulative translation adjustment created by the different rates we apply to current period income or loss and the balance sheet. For each subsidiary, we apply the monthly average functional exchange rate to its monthly income or loss and the month-end functional currency rate to translate the balance sheet.

Foreign currency transaction gains and losses are a result of the effect of exchange rate changes on transactions denominated in currencies other than the functional currency of the respective subsidiary. Transaction gains and losses are recognized in Other income (expense), net, in the consolidated statements of operations. For the years ended December 31, 2024, 2023 and 2022, we recorded a net foreign currency transaction gain of \$57 million, gain of \$122 million and loss of \$89 million, respectively.

Warranties

We provide a manufacturer's warranty on all new and used vehicles and a warranty on the installation and components of the energy generation and storage systems we sell for periods typically between 10 to 25 years. We accrue a warranty reserve for the products sold by us, which includes our best estimate of the projected costs to repair or replace items under warranties and recalls if identified. These estimates are based on actual claims incurred to date and an estimate of the nature, frequency and costs of future claims. These estimates are inherently uncertain and changes to our historical or projected warranty experience may cause material changes to the warranty reserve in the future. The warranty reserve does not include projected warranty costs associated with our vehicles subject to operating lease accounting and our solar energy systems under lease contracts or PPAs, as the costs to repair these warranty claims are expensed as incurred. The portion of the warranty reserve expected to be incurred within the next 12 months is included within Accrued liabilities and other, while the remaining balance is included within Other long-term liabilities on the consolidated balance sheets. For liabilities that we are entitled to receive indemnification from our suppliers, we record receivables for the contractually obligated amounts on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. Warranty expense is recorded as a component of Cost of revenues in the consolidated statements of operations. Due to the magnitude of our automotive business, our accrued warranty balance is primarily related to our automotive segment. Accrued warranty activity consisted of the following (in millions):

	Year Ended December 31,		
	2024	2023	2022
Accrued warranty—beginning of period	\$ 5,152	\$ 3,505	\$ 2,101
Warranty costs incurred	(1,453)	(1,225)	(803)
Net changes in liability for pre-existing warranties, including expirations and foreign exchange impact	287	539	522
Provision for warranty	2,730	2,333	1,685
Accrued warranty—end of period	<u>\$ 6,716</u>	<u>\$ 5,152</u>	<u>\$ 3,505</u>

ASU 2023-08

In December 2023, the FASB issued ASU No. 2023-08, Accounting for and Disclosure of Crypto Assets (Subtopic 350-60) (“new crypto assets standard”). The new crypto assets standard requires certain crypto assets to be measured at fair value separately on the balance sheet with changes reported in the statement of operations each reporting period. The new crypto assets standard also enhances the other intangible asset disclosure requirements by requiring the name, cost basis, fair value, and number of units for each significant crypto asset holding. The new crypto assets standard is effective for annual periods beginning after December 15, 2024, including interim periods within those fiscal years. Adoption of the new crypto assets standard requires a cumulative-effect adjustment to the opening balance of retained earnings as of the beginning of the annual reporting period in which an entity adopts the amendments. Early adoption is also permitted, including adoption in an interim period. However, if the new crypto assets standard is early adopted in an interim period, an entity must adopt the new crypto assets standard as of the beginning of the fiscal year that includes the interim period. We adopted the new crypto assets standard on a modified retrospective approach effective January 1, 2024. Refer to Note 3, *Digital Assets, Net* for the inclusion of the new required disclosures. The cumulative effect of the changes made on our January 1, 2024 consolidated balance sheet for the adoption of the new crypto assets standard was as follows (in millions):

	Balances at December 31, 2023	Adjustments from Adoption of the New Crypto Assets Standard	Balances at January 1, 2024
Assets			
Digital assets, net	\$ 184	\$ 303	\$ 487
Deferred tax assets	6,733	(67)	6,666
Stockholders' equity			
Retained earnings	27,882	236	28,118

Note 3 - Digital Assets, Net

During the years ended December 31, 2024 and 2023, we purchased and/or received immaterial amounts of digital assets. During the years ended December 31, 2023 and 2022, we recorded an immaterial amount and \$204 million of impairment losses on digital assets, respectively. The table below summarizes the amounts shown on our consolidated balance sheet as of December 31, 2024 (in millions except units of digital assets).

	December 31, 2024		
	Units	Cost Basis	Fair Value
Digital assets held:			
Bitcoin	11,509	\$ 386	\$ 1,074
Other		1	2
Total		387	1,076

The following table provides details of the activities related to our digital assets for the year ended December 31, 2024 (in millions):

Beginning balance at fair value	\$ 487
Unrealized gains, net	589
Ending balance	\$ 1,076

The following tables present summarized unaudited quarterly financial statement data for those impacted line items based on the Company’s adoption of the new crypto assets standard as described in Note 2, *Summary of Significant Accounting Policies* (in millions, except per share data):

Consolidated Balance Sheets (unaudited):	As of:		
	March 31, 2024	June 30, 2024	September 30, 2024
Digital assets, net			
Before adoption	\$ 184	\$ 184	\$ 184
Adjustments	638	538	545
As adjusted	\$ 822	\$ 722	\$ 729
Deferred tax assets			
Before adoption	\$ 6,769	\$ 6,692	\$ 6,486
Adjustments	(141)	(119)	(120)
As adjusted	\$ 6,628	\$ 6,573	\$ 6,366
Condensed Consolidated Statements of Operations (unaudited):	Three Months Ended		
	March 31, 2024	June 30, 2024	September 30, 2024
Other income (expense), net			
Before adoption	\$ 108	\$ 20	\$ (270)
Adjustments	335	(100)	7
As adjusted	\$ 443	\$ (80)	\$ (263)
Provision for (benefit from) income taxes			
Before adoption	\$ 409	\$ 393	\$ 601
Adjustments	74	(22)	1
As adjusted	\$ 483	\$ 371	\$ 602
Net income attributable to common stockholders			
Before adoption	\$ 1,129	\$ 1,478	\$ 2,167
Adjustments	261	(78)	6
As adjusted	\$ 1,390	\$ 1,400	\$ 2,173
Net income per share of common stock attributable to common stockholders			
Basic			
Before adoption	\$ 0.37	\$ 0.46	\$ 0.68
Adjustments	0.08	(0.02)	—
As adjusted	\$ 0.45	\$ 0.44	\$ 0.68
Diluted			
Before adoption	\$ 0.34	\$ 0.42	\$ 0.62
Adjustments	0.07	(0.02)	—
As adjusted	\$ 0.41	\$ 0.40	\$ 0.62

The unaudited impact of adoption for the three months ended December 31, 2024 was to increase Digital assets, net by \$347 million, with a corresponding increase in Other income (expense), net and decrease our Deferred tax assets by \$77 million with a corresponding increase in our Provision for (benefit from) income taxes, thus contributing \$270 million to Net income attributable to common stockholders. These amounts contributed \$0.08 to both basic and diluted net income attributable to common stockholders per share of common stock for the three months ended December 31, 2024.

There were no impairment losses recorded for any period during the year ended December 31, 2024. As a result, the unaudited interim balances of Digital assets, net did not change throughout the year ended December 31, 2024 prior to the adoption of the new crypto assets standard, which was adopted as of January 1, 2024.

Note 4 - Goodwill and Intangible Assets

Goodwill decreased \$9 million within the automotive segment from \$253 million as of December 31, 2023 to \$244 million as of December 31, 2024. There were no accumulated impairment losses as of December 31, 2024 and 2023.

The net carrying value of our intangible assets decreased from \$178 million as of December 31, 2023 to \$150 million as of December 31, 2024 mainly from amortization.

Note 5 - Fair Value of Financial Instruments

ASC 820 states that fair value is an exit price, representing the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants. As such, fair value is a market-based measurement that should be determined based on assumptions that market participants would use in pricing an asset or a liability. The three-tiered fair value hierarchy, which prioritizes which inputs should be used in measuring fair value, is comprised of: (Level I) observable inputs such as quoted prices in active markets; (Level II) inputs other than quoted prices in active markets that are observable either directly or indirectly and (Level III) unobservable inputs for which there is little or no market data. The fair value hierarchy requires the use of observable market data when available in determining fair value. Our assets and liabilities that were measured at fair value on a recurring basis were as follows (in millions):

	December 31, 2024				December 31, 2023			
	Fair Value	Level I	Level II	Level III	Fair Value	Level I	Level II	Level III
Certificates of deposit and time deposits	\$ 12,767	\$ —	\$ 12,767	\$ —	\$ 6,996	\$ —	\$ 6,996	\$ —
Commercial paper	3,919	—	3,919	—	470	—	470	—
U.S. government securities	3,620	—	3,620	—	5,136	—	5,136	—
Corporate debt securities	118	—	118	—	480	—	480	—
Money market funds	1,753	1,753	—	—	109	109	—	—
Digital assets	1,076	1,076	—	—	Not applicable			
Total	\$ 23,253	\$ 2,829	\$ 20,424	\$ —	\$ 13,191	\$ 109	\$ 13,082	\$ —

All of our money market funds and digital assets are classified within Level I of the fair value hierarchy because they were valued using quoted prices in active markets. Our U.S. government securities, certificates of deposit, commercial paper, time deposits and corporate debt securities are classified within Level II of the fair value hierarchy and the market approach was used to determine fair value of these investments.

Our cash, cash equivalents and investments classified by security type as of December 31, 2024 and 2023 consisted of the following (in millions):

	December 31, 2024					
	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Short-Term Investments
Cash	\$ 14,386	\$ —	\$ —	\$ 14,386	\$ 14,386	\$ —
Certificates of deposit and time deposits	12,767	—	—	12,767	—	12,767
Commercial paper	3,908	11	—	3,919	—	3,919
U.S. government securities	3,618	3	(1)	3,620	—	3,620
Corporate debt securities	117	1	—	118	—	118
Money market funds	1,753	—	—	1,753	1,753	—
Total cash, cash equivalents and short-term investments	\$ 36,549	\$ 15	\$ (1)	\$ 36,563	\$ 16,139	\$ 20,424

	December 31, 2023					
	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Short-Term Investments
Cash	\$ 15,903	\$ —	\$ —	\$ 15,903	\$ 15,903	\$ —
Certificates of deposit and time deposits	6,995	1	—	6,996	—	6,996
U.S. government securities	5,136	1	(1)	5,136	277	4,859
Corporate debt securities	485	1	(6)	480	—	480
Commercial paper	470	—	—	470	109	361
Money market funds	109	—	—	109	109	—
Total cash, cash equivalents and short-term investments	\$ 29,098	\$ 3	\$ (7)	\$ 29,094	\$ 16,398	\$ 12,696

As of December 31, 2024, the vast majority of our short-term investments had contractual maturity dates within one year.

Disclosure of Fair Values

Our financial instruments that are not re-measured at fair value include accounts receivable, financing receivables, other receivables, accounts payable, accrued liabilities, customer deposits and debt. The carrying values of these financial instruments materially approximate their fair values.

Note 6 - Inventory

Our inventory consisted of the following (in millions):

	December 31, 2024	December 31, 2023
Raw materials	\$ 5,242	\$ 5,390
Work in process	1,532	2,016
Finished goods (1)	3,940	5,049
Service parts	1,303	1,171
Total	\$ 12,017	\$ 13,626

(1) Finished goods inventory includes products-in-transit to fulfill customer orders, new vehicles, used vehicles and energy products available for sale.

We write-down inventory for any excess or obsolete inventory or when we believe that the net realizable value of inventory is less than the carrying value. During the years ended December 31, 2024, 2023 and 2022, we recorded write-downs of \$190 million, \$233 million and \$144 million, respectively, in Cost of revenues in the consolidated statements of operations.